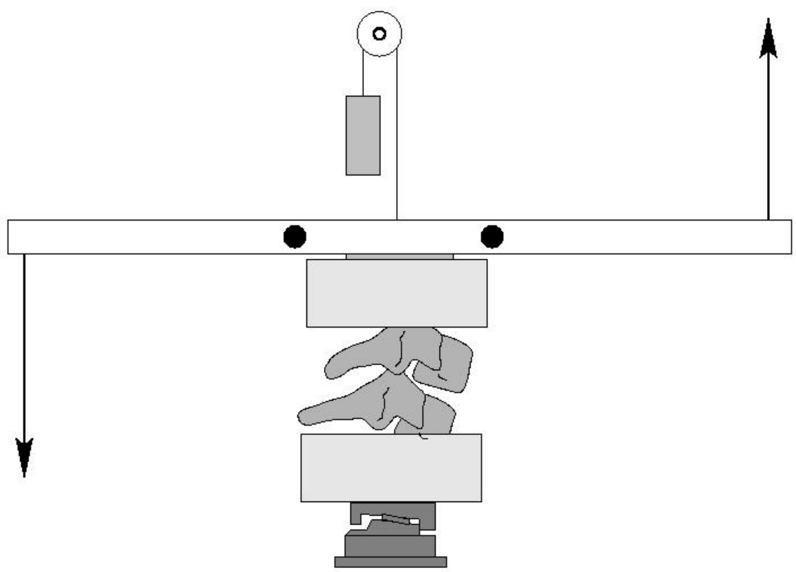
Bending Responses Of The Human Cervical Spine

Flexibility Test Report

Test Reference: B03F34FLEX

Test Date: 3/25/98 Submission Date: 8/15/00



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NHTSA Cooperative Agreement No. *DTNH22-94-Y-07133*

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Study Objective: The objective of this study is to provide data on the responses of the human cervical spine to pure bending. The experimental protocol consists of two tests: a flexibility test and a failure test. The flexibility test is designed to quantify the static response of motion segments to flexion and extension moments. The failure test is designed to quantify the bending tolerance of motion segments. See the Methods Section for more details on the experimental protocol.

Test Objective: Flexibility Test

Test Reference: *B03F34FLEX*

Motion Segment: C3 - C4

Specimen ID: *B03F*

Specimen Description

Sex: *Female* Age: *51 years*

Weight: *556 (N)* Height: *158 (cm)*

Comments: The specimen was in good condition. There was a small osteophyte at C6.

Results:

Sign Conventions: Compliant with SAE J211/1 March 1995

Load Cell	Measure	Manipulation	Polarity
GSE 6607 Upper Neck Load Cell	Fx	Head Rearward	+
	Fy	Head Leftward	+
	Fz	Head Upward	+
	Mx	Left Ear to Left Shoulder	+
	My	Chin to Sternum	+

Data Channels

Curve	Measurement	Location	Axis	Unit	Min	Max	Comments
1	Computed	NEKL	YL	NWM	-4.3	4.1	MOMENT-ANGLE

All channels have had small biases removed. The biases were recorded when the channels were zeroed during the establishment of the reference position. See the Appendix for more details. Only the My moment channel was recorded for each load step. The resulting angular displacements of the motion segments were calculated from digital images.

Image Data:

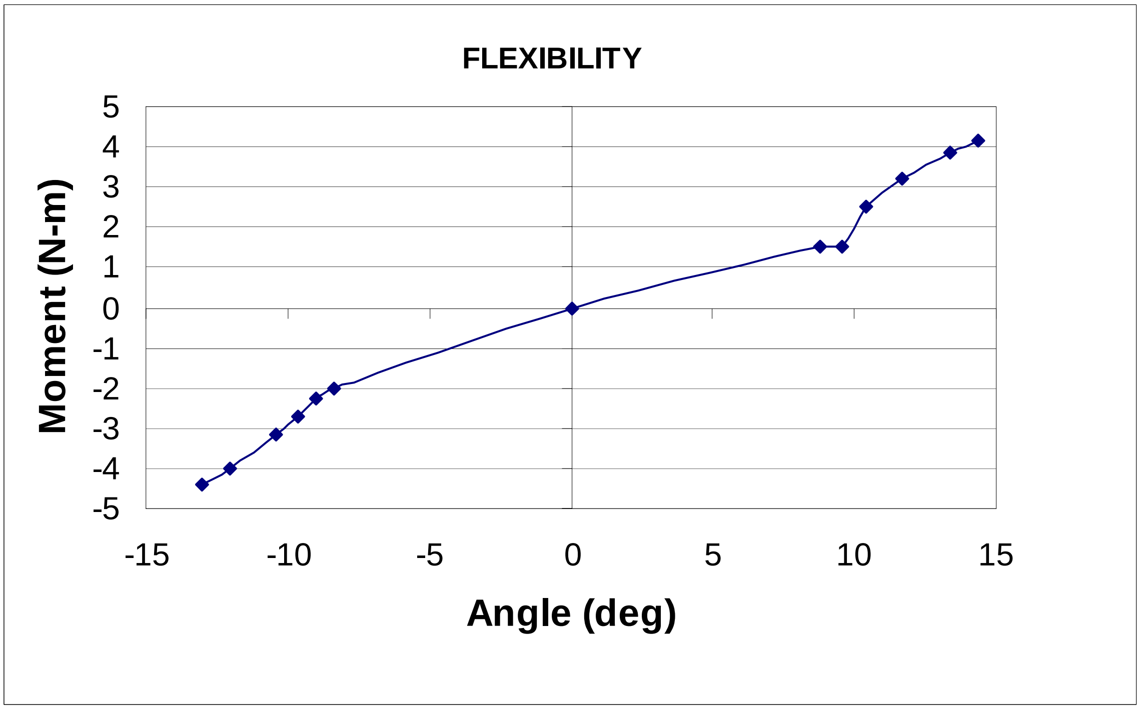
Reference Image: ***F07.tif***

The digital images are of sequential moment load steps for flexion through extension. The angular displacements of the spinal segments are computed relative to the Reference Image. This Reference Image is also used for calculating angular displacements in the failure test.

Comments:

Computed Data

B03F34FLEX.1



Methods

Unembalmed human heads with intact spines were obtained shortly after death. All specimen handling was performed in compliance with both NHTSA 700-4 guidelines and CDC guidelines. The specimens were sealed in plastic bags, frozen and stored at -20°. All donor medical records were examined to ensure that there were no pre-existing pathologies, such as severe degenerative diseases or spinal pathologies, which could affect the structural responses of the specimens.

The muscular tissues were removed while keeping all the ligamentous structures intact. The lower cervical motion segments (C3-C4, C5-C6, C7-T1) were isolated, cleaned, and cast into aluminum cups with fiber-reinforced polyester resin. The cephalad end of the upper cervical spine specimens was secured using halo fixation. If necessary, the mandible and maxilla were removed in order to allow appropriate flexion range of motion. Upper cervical specimens (Head-C2) were inverted and mounted in the test frame using halo fixation of the head and casting of the C2 vertebra.

The test apparatus applies pure bending moments, and the resulting angular displacements are computed by tracking markers on digital images. The moments are generated using a pair of pneumatic pistons to apply a force-couple. After the specimen was mounted in the test frame, a counterweight was applied to the upper casting assembly. The mass of the counterweight was chosen to counterbalance the mass of the assembly, and to place the motion segment in 0.5 Newtons of tension. The purpose of the 0.5 N tensile load was to establish an initial position for the motion segment in the middle of it's neutral zone. The specimen was imaged in this position and the load cell values were zeroed. This image served as the reference position for the flexibility and the failure tests. Prior to flexibility testing, the specimens were preconditioned with 30 cycles of ±1.5 N-m of moment. The specimens were then loaded with pure flexion and extension moments in 0.5 N-m increments. Thirty seconds of creep was allowed prior to data acquisition, and the load was released between load steps. The peak applied moment was approximately 3.5 N-m. A six-axis load cell (GSE model 6607-00) at the base of the specimen was used to ensure that the moment remained pure. The load cell is also the source of the moment data.

After completion of the above bending tests, the tension mass was removed, and the specimens were failed in either flexion or extension. The loading rate was the maximum that could be achieved using our 100 psi system. The rates were dependent on the stiffness of the specimen, and were on the order of 90 N-m/second. The failures tests were imaged at 125 frames/second. Failure was defined as a decrease in the measured moment with increasing rotation, and they were verified by examination of the high speed images. Specimen dissection was performed to document injuries.

All the image data was uploaded to an SGI running a modified version of the xv shareware package. This software enables us to digitize the location of marker pins in each image. The location of each marker is determined using a centroid algorithm to increase accuracy. Angular displacements for each image are determined with respect to the reference image.